Logic Gates & Boolean Algebra

1. Classification of Logic Gates

Theory:

Logic gates are the physical hardware building blocks of digital circuits. They can be broadly classified into three categories, and circuits built from them are either **Combinational** (output depends only on current input) or **Sequential** (output depends on current input and past states/memory).

- **Basic Gates:** AND, OR, NOT
- **Universal Gates:** NAND, NOR (Called "universal" because you can build any other gate using only these).
- **Arithmetic / Derived Gates:** XOR (Exclusive-OR), XNOR (Exclusive-NOR).

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2. Logic Gate Truth Tables

Theory:

A truth table defines the output behavior of a logic gate for every possible combination of inputs. In Boolean addition, remember that $1 + 1 = 1$ (because 1 represents "True" or "High voltage", and True + True is still True).

Basic & Universal Gates:

Inputs		AND	OR	NAND	NOR
A	B	$y = A \cdot B$	$y = A + B$	$y = \overline{A \cdot B}$	$y = \overline{A + B}$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

Arithmetic Gates:

Inputs		XOR	XNOR
A	B	$y = A \oplus B$	$y = \overline{A \oplus B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

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3. Boolean Algebra Fundamentals

Theory:

Boolean algebra is the mathematics of digital logic. Instead of standard numbers, it uses binary variables to represent logical states.

- **Variable:** A symbol used to represent a logical quantity (e.g., A, B, C, x, y, z).
- **Complement:** The exact inverse of a variable, denoted by a bar over the letter (e.g., $\overline{A}, \overline{B}, \overline{C}$).
- **Literal:** A variable or its complement (both A and $\overline{A}$ are literals).

The Three Basic Laws

$$\begin{aligned} \text{Commutative Law:} \quad & A + B = B + A \\ & A \cdot B = B \cdot A \end{aligned}$$

$$\begin{aligned} \text{Associative Law:} \quad & A + (B + C) = (A + B) + C \\ & A \cdot (B \cdot C) = (A \cdot B) \cdot C \end{aligned}$$

$$\begin{aligned} \text{Distributive Law:} \quad & X \cdot (Y + Z) = XY + XZ \\ & A + BC = (A + B)(A + C) \end{aligned}$$

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4. Boolean Rules & Identities

Theory:

These 12 rules are the shortcuts you need to simplify complex digital circuits. Memorizing these will save you massive amounts of time when minimizing logic expressions.

1.	$A + 0 = A$	7.	$A \cdot A = A$
2.	$A + 1 = 1$	8.	$A \cdot \overline{A} = 0$
3.	$A \cdot 0 = 0$	9.	$\overline{\overline{A}} = A$
4.	$A \cdot 1 = A$	10.	$A + AB = A$
5.	$A + A = A$	11.	$A + \overline{A}B = A + B$
6.	$A + \overline{A} = 1$	12.	$(A + B)(A + C) = A + BC$

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5. Boolean Proofs

Theory:

We can prove complex Boolean rules by breaking them down algebraically using the basic identities listed above.

(Proof 1: Distributive Rule Expansion)

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\begin{aligned}
(A + B)(A + C) &= AA + AC + AB + BC && \text{(Standard multiplication)} \\
&= A + AC + AB + BC && \text{(Apply Rule 7: } AA = A\text{)} \\
&= A(1 + C) + AB + BC && \text{(Factor out } A\text{)} \\
&= A(1) + AB + BC && \text{(Apply Rule 2: } 1 + C = 1\text{)} \\
&= A(1 + B) + BC && \text{(Factor out } A \text{ again)} \\
&= A(1) + BC && \text{(Apply Rule 2: } 1 + B = 1\text{)} \\
&= A + BC && \textbf{(Proved)}
\end{aligned}
$$

(Proof 2: Rule 11)

$$
\begin{aligned}
A + \overline{A}B &= A \cdot A + AB + \overline{A}B && \text{(Expand } A \text{ into } A \cdot A + AB\text{)} \\
&= A \cdot A + AB + A\overline{A} + \overline{A}B && \text{(Add } A\overline{A}\text{, which equals 0)} \\
&= A(A + B) + \overline{A}(A + B) && \text{(Factor } A \text{ from first pair, } \overline{A} \text{ from second)} \\
&= (A + \overline{A})(A + B) && \text{(Combine terms)} \\
&= 1 \cdot (A + B) && \text{(Apply Rule 6: } A + \overline{A} = 1\text{)} \\
&= A + B && \textbf{(Proved)}
\end{aligned}
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